Assignment 8: Application Optimization, Scalability and Reliability

ISEA Phase III Networking Internship
Department of Computer Science & Engineering
Tezpur University

Total Marks: 100

Objective

Enhance the existing GUI-based multi-client chat application by improving scalability, reliability, maintainability and overall software quality. Build upon Assignment 7 instead of creating a new application.

Learning Outcomes

- Improve scalability and reliability.
- Optimize an existing client-server application.
- Evaluate performance before and after optimization.
- Improve resource management and exception handling.

Prerequisite

Continue from Assignment 7. Starting from scratch is not permitted.

Network Setup

Use Mininet:

sudo mn --topo single,11

Test using 5, 8 and 10 concurrent clients.

Development Guidelines

- Reuse the existing client and server.
- Do not redesign the communication protocol.
- Focus on software quality improvements.
- Document every optimization implemented.

Task 1: Connection Management (15 Marks)

Detect disconnected clients automatically, remove inactive clients, release resources correctly and display meaningful error messages.

Task 2: Reliability Enhancement (15 Marks)

Implement automatic reconnection, graceful shutdown, timeout handling and better exception handling.

Task 3: Scalability Enhancement (20 Marks)

Support at least 10 concurrent clients without crashes. Improve thread management or implement another scalability enhancement.

Task 4: Configuration Management (10 Marks)

Move configurable values into config.json and avoid hardcoded parameters.

Task 5: Performance Evaluation (15 Marks)

Compare performance before and after optimization. Measure delay, throughput, CPU usage and memory usage. Save results in performance_results.csv and generate graphs.

Task 6: Wireshark Verification (5 Marks)

Capture and explain TCP communication during normal operation.

Task 7: GitHub Update (5 Marks)

Update the existing repository. Continue using the naming convention beginning with ISEA-Phase3-TezpurUniversity-.

Task 8: Handwritten Reflection (5 Marks)

Answer the reflection questions by hand, scan them and submit them with the report.

Handwritten Reflection Questions

1. Which optimization produced the greatest improvement?
2. Why is scalability important?
3. What reliability issues did you discover?
4. Which optimization was the most difficult?
5. What additional improvements are required to support 100 users?

Required Outputs

- server.py
- client_gui.py
- config.json
- performance_results.csv
- graphs/
- screenshots/
- report.pdf
- handwritten_reflection.pdf

Report Format

6. Objective
7. Scalability Improvements
8. Reliability Improvements
9. Configuration Management
10. Experimental Setup

11. Performance Results
12. Graph Analysis
13. Wireshark Verification
14. Challenges Faced
15. Conclusion

Evaluation Rubric

Component	Marks
Connection Management	15
Reliability Enhancement	15
Scalability Enhancement	20
Configuration Management	10
Performance Evaluation	15
Wireshark Verification	5
GitHub Update	5
Handwritten Reflection	5
Report Quality	10
Total	100

Important Notes

- Reuse Assignment 7 implementation.
- Do not build a new application.
- Generate all graphs from actual experiments.
- Be prepared to demonstrate the application.
- Handwritten reflections are mandatory.